

UIKit and Slint: Scroll Physics

A physical comparison of pan entry, overscroll pulling, and return animation, with a simulator follow-up.

Measured progress: the pull mapping matches the three tested exposure peaks closely. The changed spring is much closer, but still does not match the full UIKit curve.

0.069 pt

Pull peak gap at 200 points

40.18 to 7.76 pt

600-point return gap, before / candidate

NOT MATCHED

Spring criterion still fails

01 Before changes - Original behavior, event dispatch, and a separate manual recording.

02 Fixing the pull - A 10-point threshold and absolute rubber-band mapping, tested in code.

03 The return animation - Candidate spring, release-state controls, and natural collisions after finger lift.

Murmele/slnt, branch nigel/ios-uikit-scroll-parity. Draft PR #7. No chart rescales distance or duration. Raw per-event recordings remain local.

Share the current work: github.com/Murmele/slnt/pull/7

The original behavior and the measurement

The app overlays a translucent native UIScrollView on a Slint ScrollView and forwards the original touch objects and event to Slint. Both views have matching 408 x 774-point viewports and 400 x 72000-point content, with 72-point rows.

The physical iPhone 13 Pro Max runs iOS 27.0. The build uses Release, Cupertino style, and Winit/Skia. The automated position sampler delivered about 120 Hz: median intervals near 8.335 ms. A point is three physical pixels on this phone. Sampling rate is not renderer FPS.

Original Slint subtracts 8 points and applies overscroll friction to each event delta. The current exposure affects the next delta, and a first boundary crossing can pass through unresisted. Event batching can therefore change the retained exposure.

Original 120 Hz capture	UIKit peak exposure	Slint peak exposure
50-point pull	21.333 pt	21.567 pt
200-point pull	92.000 pt	89.352 pt
600-point pull	228.667 pt	221.033 pt

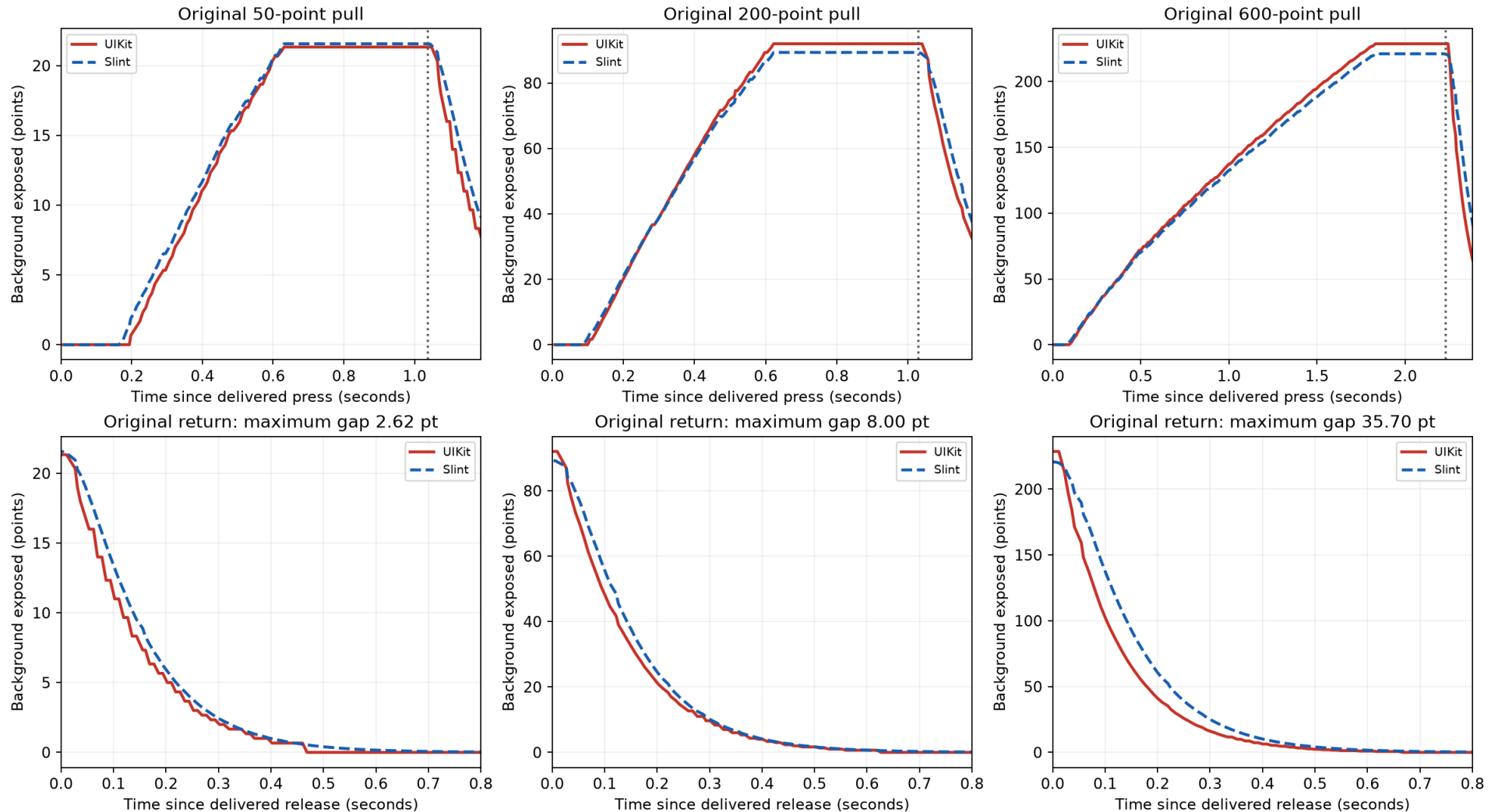
UIKit also discarded more initial movement. Held paths consumed 10 points in UIKit and 8 in Slint; immediate earlier paths consumed 10-15.333 points in UIKit. That is discarded movement, not proof of a universal recognition threshold.

Recognition and action dispatch are distinct. Apple documents the began action on the next run-loop cycle. Disabling forwarding to Slint did not remove the extra loss, although recording and scheduling effects are not completely excluded.

Original automated pull and return

Actual position samples from the physical phone, paired within each capture.

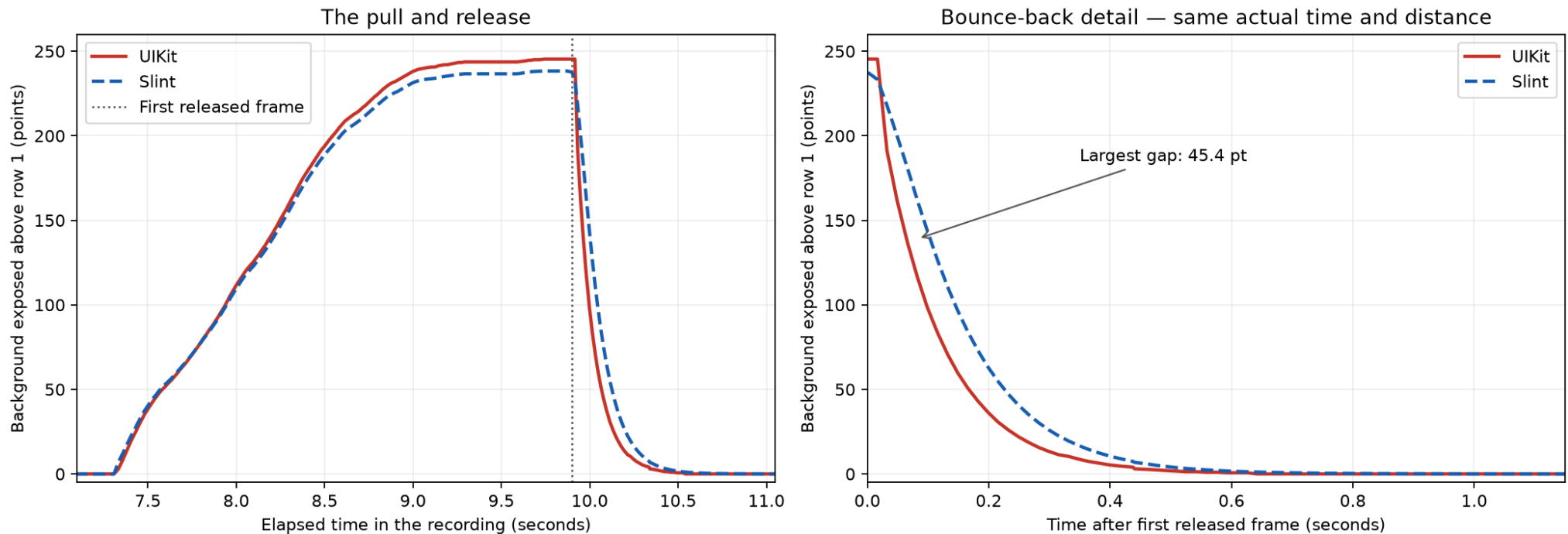
Part 1: original Slint versus UIKit on the physical iPhone



A separate real-finger recording

Earlier 60 Hz manual capture: 45.418-point largest gap; UIKit / Slint settling about 0.642 / 0.742 s.

Your iPhone overscroll recording — UIKit vs Slint
One manual pull; no distance or duration normalization



From event-by-event friction to a distance map

The pull candidate uses 10 points for the iOS threshold and maps the underlying drag distance into visible exposure. Each event inverts the current exposure, adds the new finger delta, then applies the same map. Large batches and small steps have the same mathematical result.

exposure = 0.55 x distance x viewport / (viewport + 0.55 x distance)

The coefficient was proposed from the 200-point capture. The 50- and 600-point pulls were checks outside that calibration distance. No mode uses UIKit positions to drive Slint.

Delivered pull	UIKit	Modified Slint	Absolute peak gap
50 pt	21.333 pt	21.392 pt	0.059 pt
200 pt	92.000 pt	92.069 pt	0.069 pt
600 pt	228.667 pt	228.642 pt	0.025 pt

A unit test checks batching independence and reversals at both bounds. On the explicitly held 40-point pan, the threshold change removed the 2-point gap in both repeats, including zero sampled contact separation.

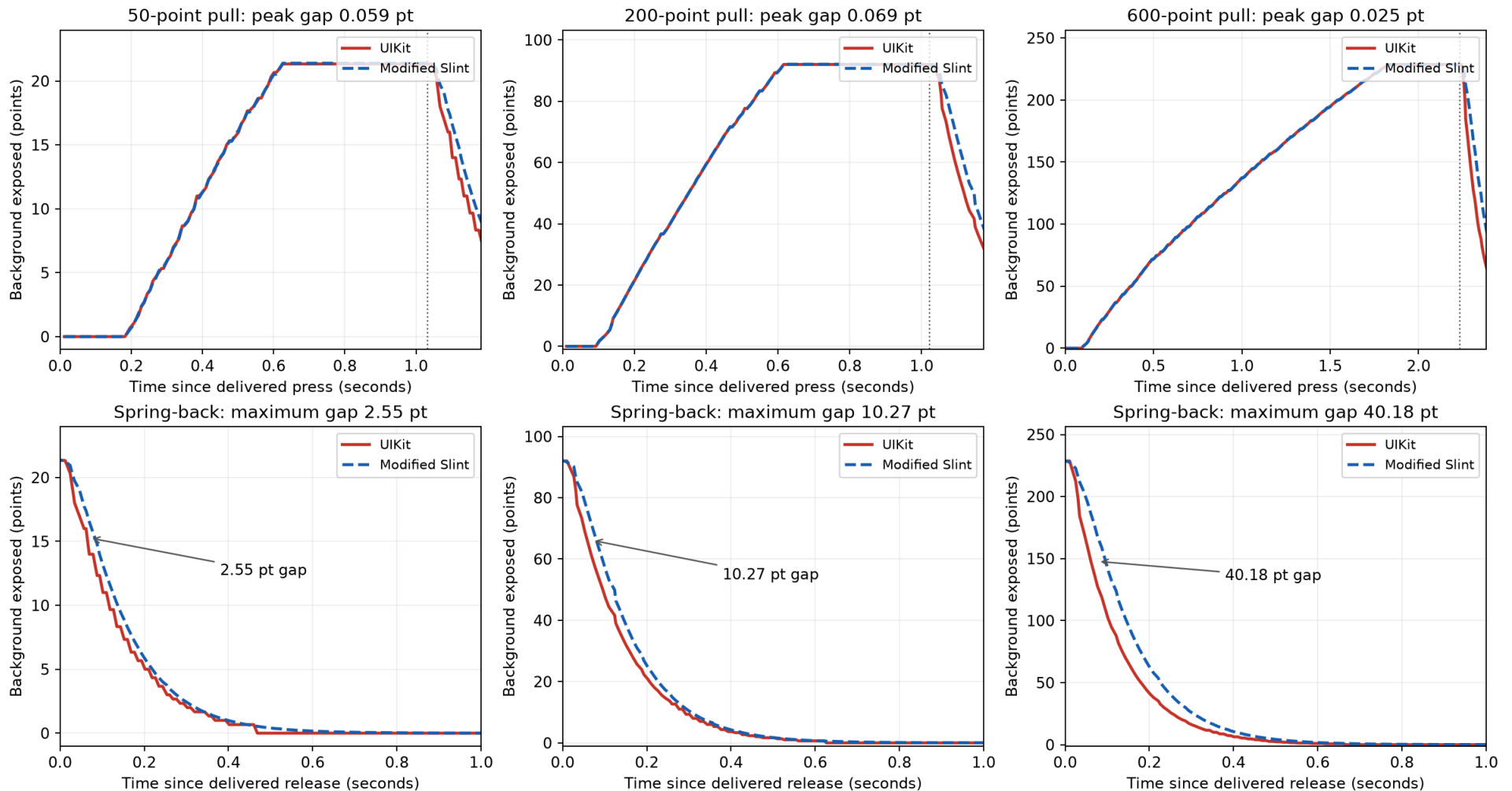
Immediate paths still diverged. A numeric threshold change alone does not reproduce UIKit's deferred baseline handling. Each chart panel compares its own paired UIKit and Slint data; requested waypoints are not delivered event counts.

Improved exposure; original spring still diverges

The pull mapping is closely matched. These captures still use the original return spring.

New physical iPhone tests: UIKit vs modified Slint

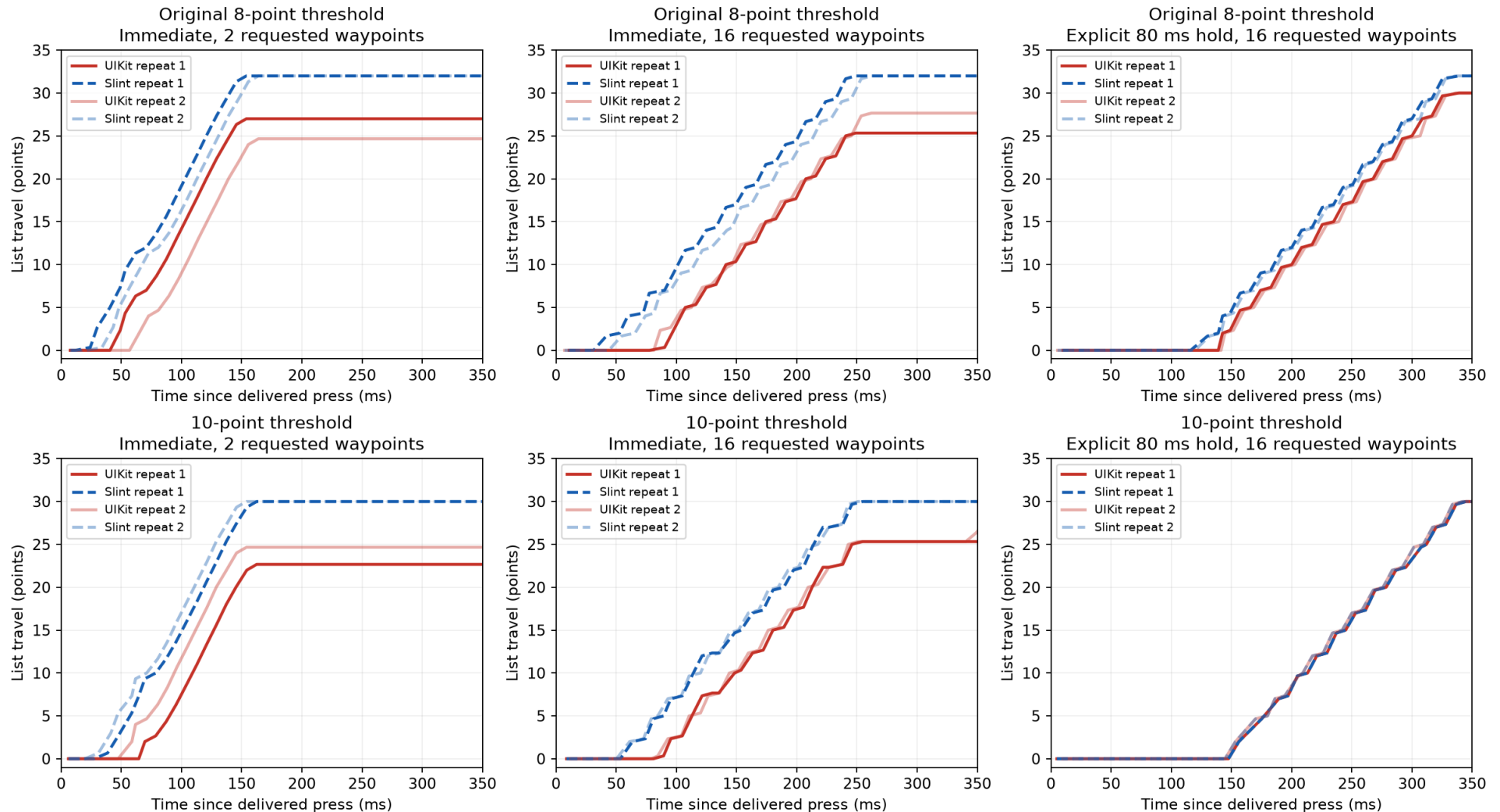
10-point threshold + absolute rubber-band mapping; release spring unchanged



Threshold change: two repeated captures

Held entry aligns here; immediate entry remains different. Every panel contains both implementations.

Part 2: threshold code change, two repeated captures per condition



A closer spring, with visible failed predictions

The original spring evolves the displayed exposure directly, with mass 0.5, stiffness 100, damping ratio 1.1, and zero initial velocity. Matching the pull leaves a large return error.

The candidate evolves spring travel in the underlying drag coordinate and maps that travel back to the display. It retains the mass, stiffness, and damping parameters. An empirical initial-return rate of 2.4422646 per second was calibrated from the 200-point trace.

Delivered pull	Original spring maximum gap	Candidate maximum gap
50 pt	2.549 pt	1.167 pt
200 pt	10.269 pt	2.458 pt
600 pt	40.177 pt	7.760 pt

The model is a candidate black-box approximation, not a claim about UIKit private code. Unseen 100-, 300-, and 450-point paths were each repeated twice. They retained maximum errors of 1.20-1.89, 3.76-4.02, and 6.97-7.20 points respectively.

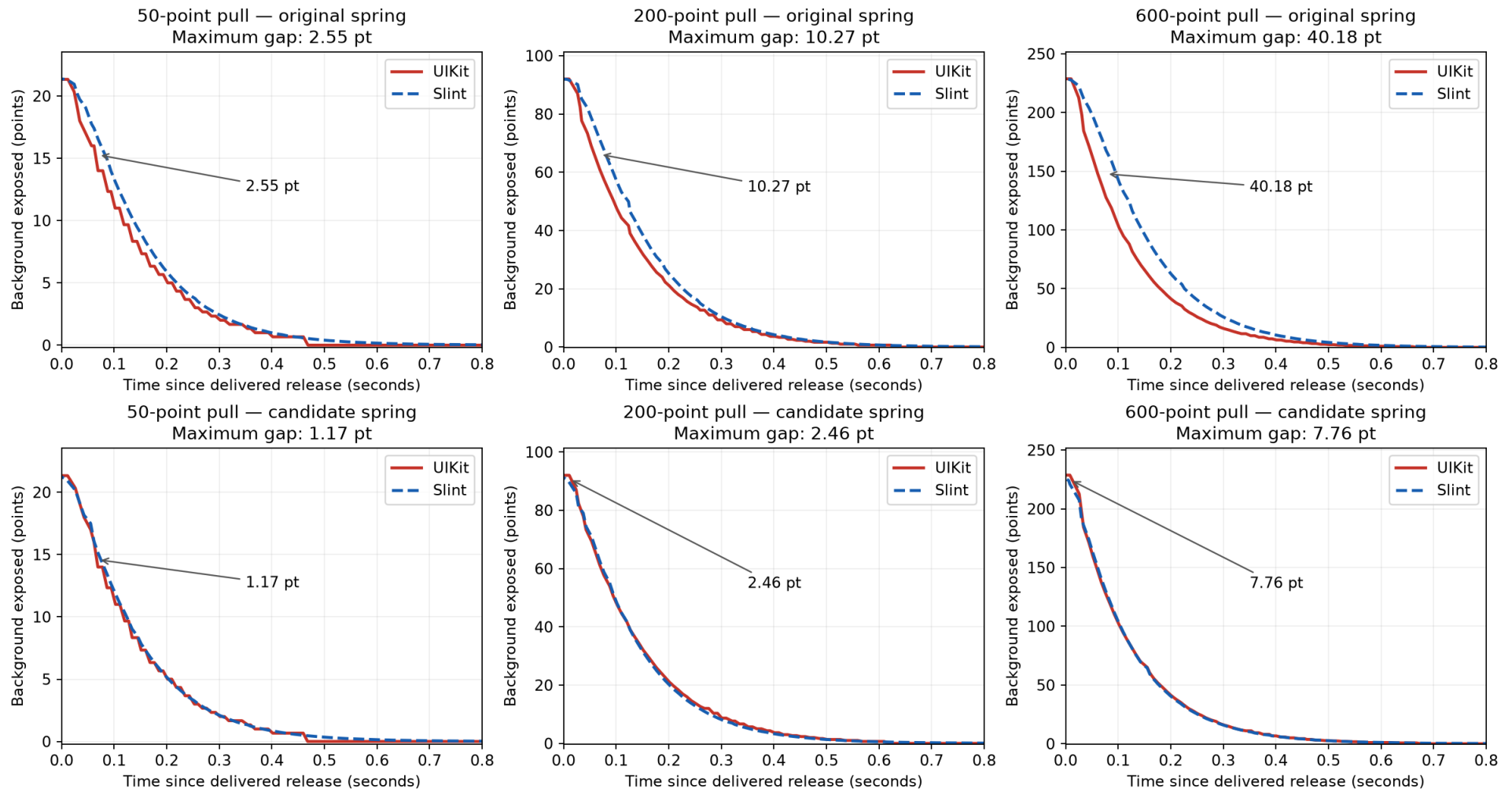
The criterion still fails: at most 0.5 points throughout return in every complete trace. The spring is improved, not solved. The early-frame residual remains important even when the later curves look close.

The final clock control starts the same model from live backend time. It worsened return RMS error in both repeats at both tested distances and is retained as a rejected control.

Changed code: return before and after

Both rows use the improved pull mapping. The remaining candidate gap is shown at actual scale.

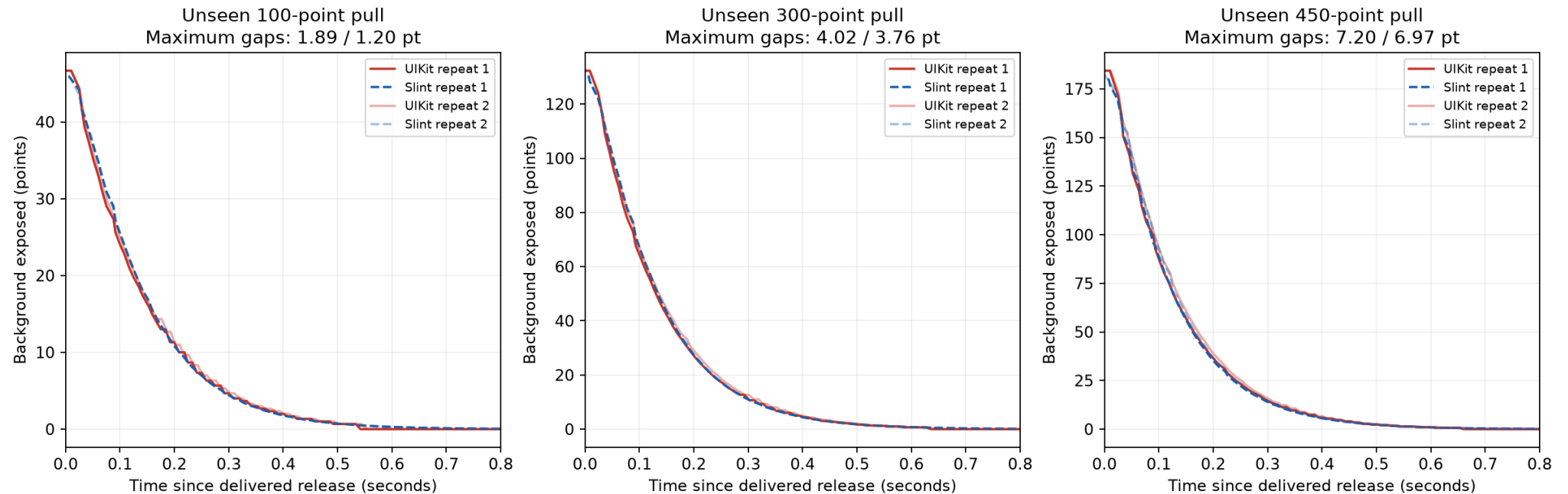
Physical iPhone spring-back: original versus changed Slint code
Both rows use the improved pull mapping; no time or distance normalization



Independent validation paths

Three distances outside the calibration path, repeated twice. Failures remain in the published aggregates.

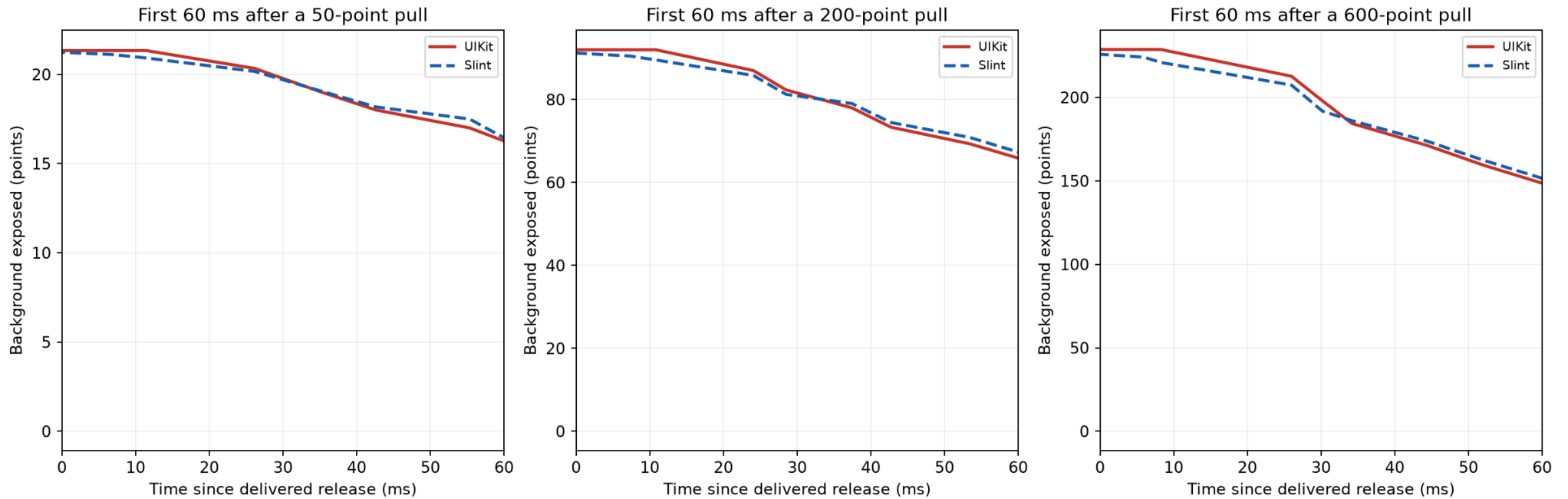
Candidate spring: independent physical iPhone validation paths



Zoom into the early release residual

Actual first 60 ms; the axes are not normalized or shifted to force alignment.

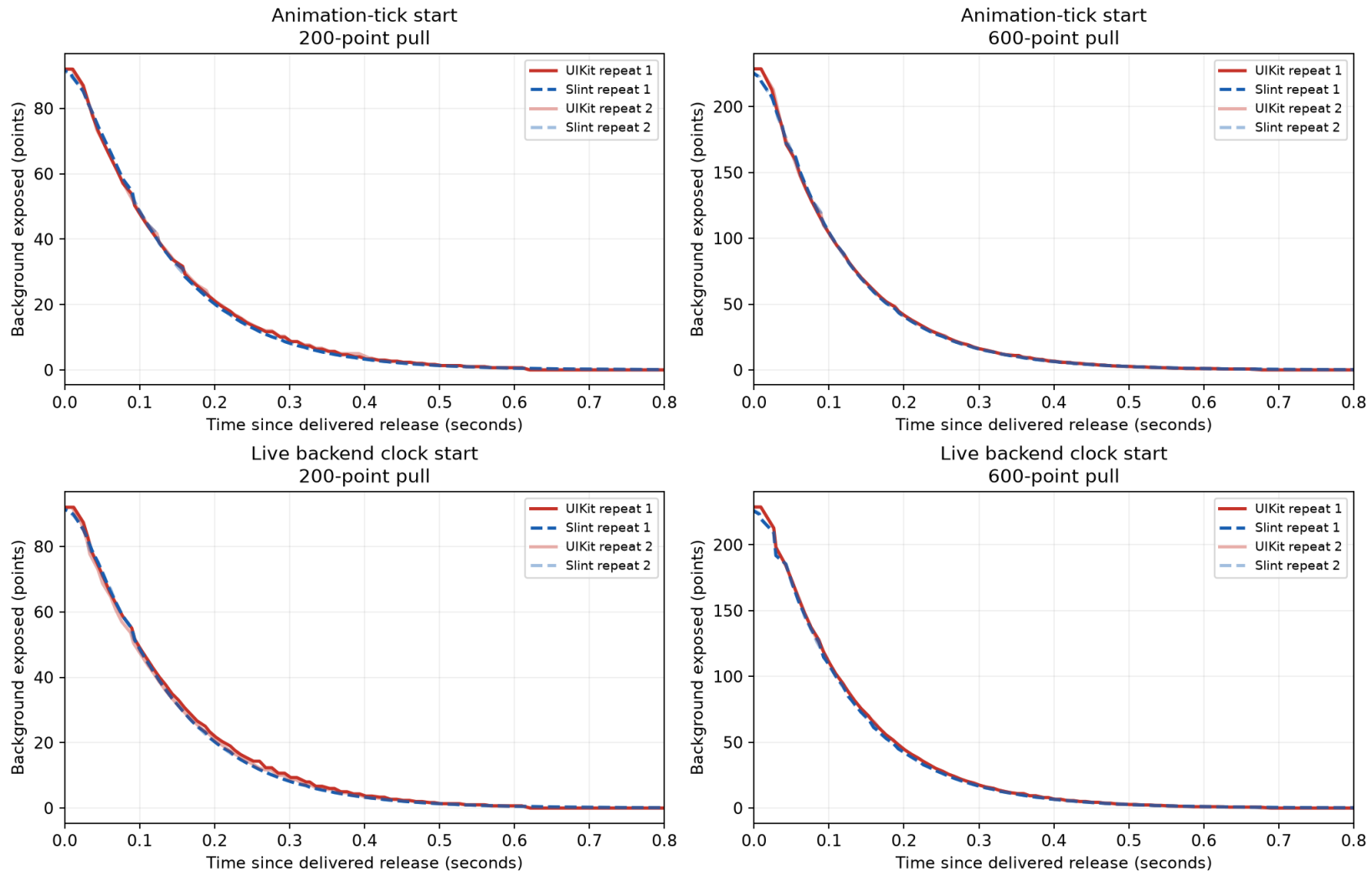
Part 3: early release residuals remain visible at actual scale



Phone follow-up: testing the release clock

Animation-tick start versus live backend clock start, each paired with UIKit and repeated twice.

Part 3: the final clock experiment did not improve return RMS error



What remains needed for a complete match

Fixed 0-0.8 s return window	Animation-tick RMS, repeats 1 / 2	Live-clock RMS, repeats 1 / 2
200-point pull	0.712 / 0.771 pt	1.014 / 0.784 pt
600-point pull	1.481 / 1.682 pt	1.864 / 1.818 pt

The recorder saw 400-409 ms of clock age before forwarding release after the hold, but only 1.375-2.724 ms at UIKit's ended action after processing. These are different dispatch observations, not clock age at spring construction. They do not support a 400 ms stale clock explanation.

Remaining work: reproduce recognized pan dispatch and the first retained movement; separate animation handoff from display-phase timing; preserve measured release state in the spring; validate bottom bounds, interrupted springs, and carried momentum independently. Native pan state, translation, and velocity are available from UIPanGestureRecognizer, but that backend approach has not been verified here.

Original two rounds: 44 complete automated captures, plus the separate manual capture. Delivered distances, explicit initial holds, and actual sampling intervals were checked. Five spring unit tests and the rubber-band batching/reversal test passed. Green UI tests mean both views received the gesture and settled, not that physics match.

The repository includes the source app, selected tests, figures, written predictions, aggregate CSV/JSON results, and plotting code. Raw per-event touch recordings and xcresult bundles remain local. The first inconsistent threshold attempt is preserved locally and excluded from the reported successful code comparisons.

Read FINDINGS.md and README.md in tests/manual/ios-scroll-physics-comparison. The report cites Apple documentation for pan states, translation, velocity, and deferred began dispatch. No Flutter claim is used as evidence of UIKit parity.

New alternatives without using the phone

The iPhone 18 Pro Simulator runs iOS 27.0 in Release with the same Cupertino/Skia app. Its viewport is 382 x 722 points and the position sampler delivers about 60 Hz. Each alternative has two 200-point and two 600-point pulls. The simulator is compared against its own UIKit reference, not against the phone curves.

Spring / delivery mode	200-point RMS, repeats 1 / 2	600-point RMS, repeats 1 / 2
Current candidate	1.870 / 3.073 pt	9.503 / 9.860 pt
Deferred release	Invalid release	Invalid release
Zero initial spring velocity	3.899 / 1.025 pt	2.826 / 2.763 pt
Direct touch time/history	2.722 / 3.125 pt	9.669 / 11.440 pt

Qualification: 11 of 16 traces have a complete gesture and settled return, the requested distance delivered within 0.5 points, and no return sample gap above 30 ms. 0 traces meet the 0.5-point full-return criterion. Every graph uses actual release callback time and unscaled points.

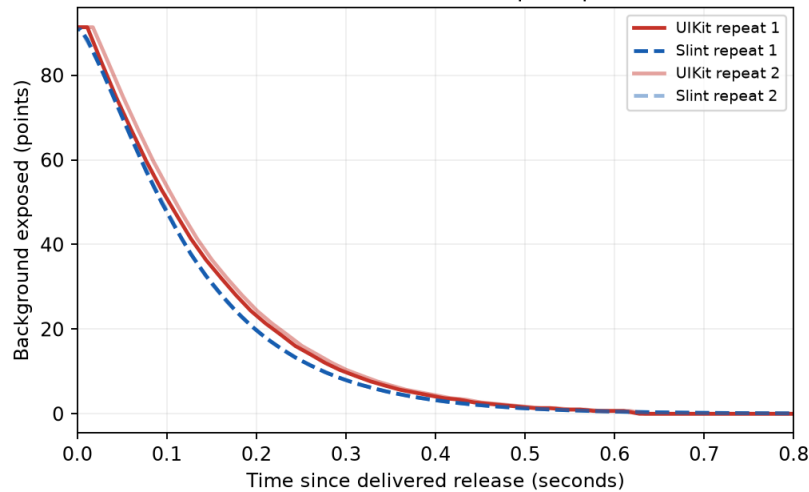
The deferred-release mode lost Slint release delivery in all four traces; its flat curves are invalid for spring comparison. Zero velocity tests the fitted initial spring kick. Direct touch delivery preserves UIKit sample times and coalesced positions that the Winit adapter drops. It also refreshes timers and changes dispatch route, so a route control is required before attributing any difference to history alone.

The core still drops the release timestamp when it constructs the pointer release. These diagnostic modes do not change normal defaults and do not establish physical-phone parity. All predictions and per-gesture aggregates are retained, including failures.

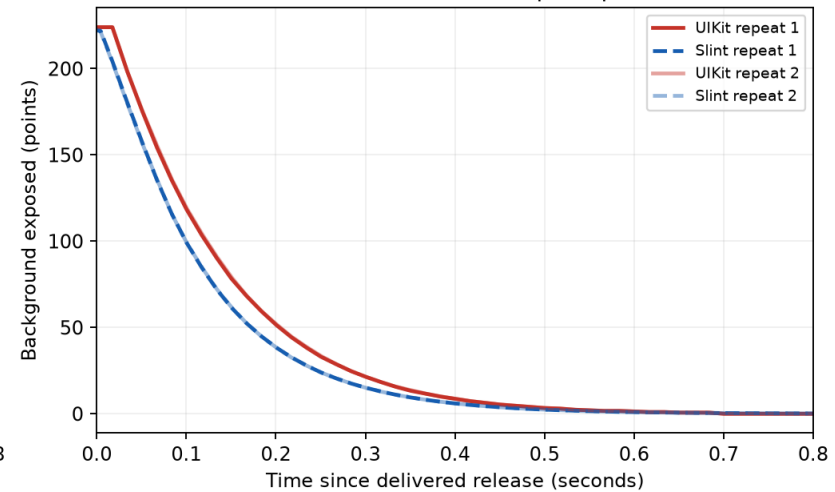
Test release ordering against the current spring

Two repeats at each distance, each with its own UIKit reference. No time alignment shift.

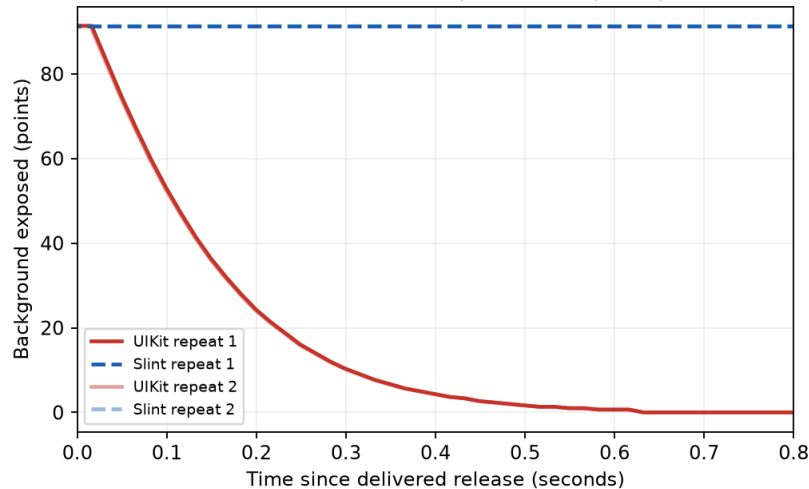
Simulator: current spring and deferred release, actual time and distance
Current candidate: 200-point pull



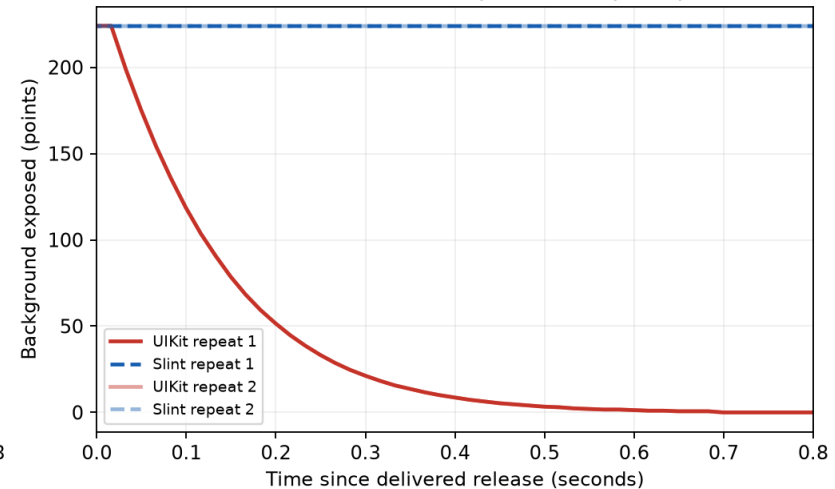
Current candidate: 600-point pull



Defer release one run-loop turn: 200-point pull



Defer release one run-loop turn: 600-point pull

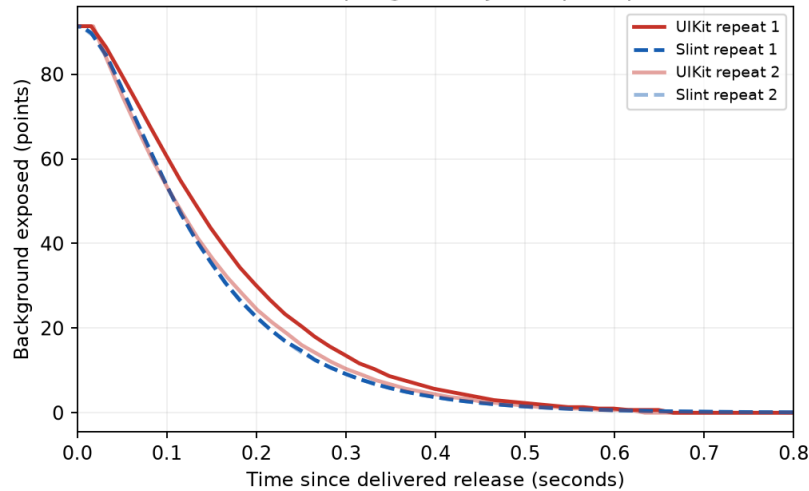


Test initial velocity and preserved touch samples

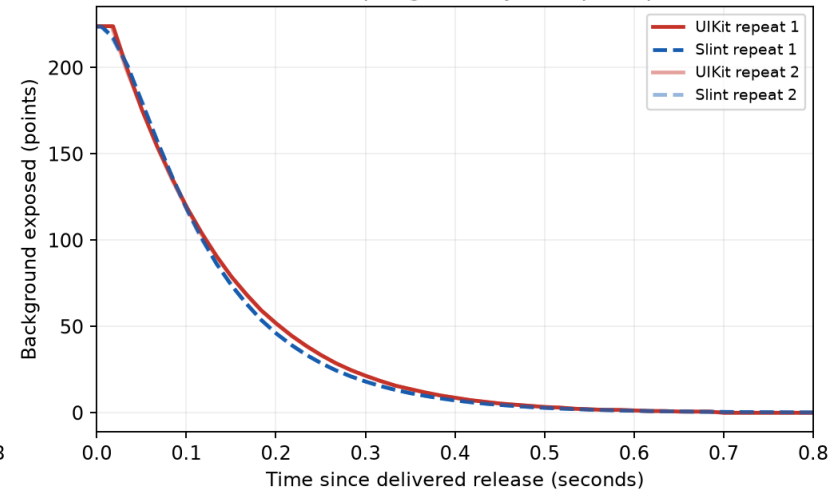
Zero spring kick and direct touch history are independent diagnostics. Both list positions appear in every panel.

Simulator: zero initial velocity and direct touch history, actual time and distance

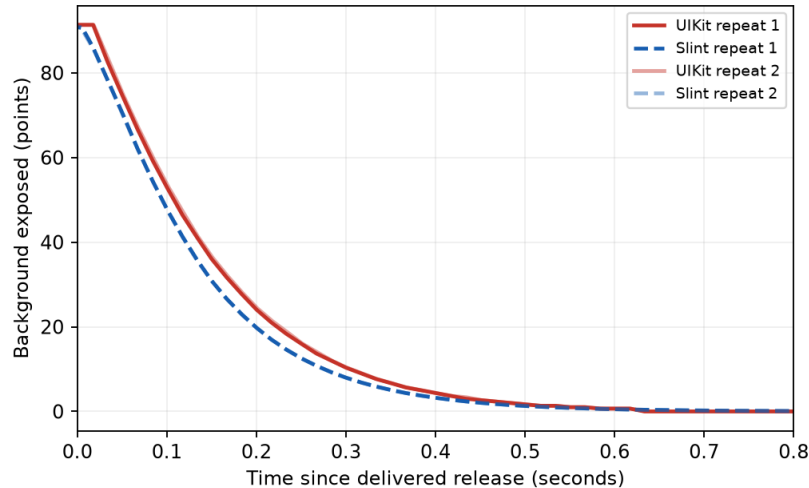
Zero initial spring velocity: 200-point pull



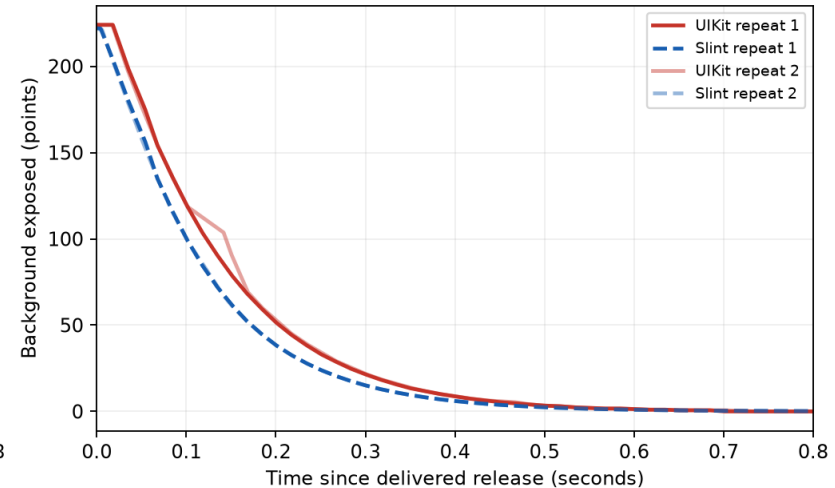
Zero initial spring velocity: 600-point pull



Direct touch time and history: 200-point pull



Direct touch time and history: 600-point pull



Immediate release, exact hold, and quiet stop

New captures use 100- and 600-point pulls, two velocity settings (400 and 1600), immediate release or a delivered 400 ms hold, and two repeats on each platform. Velocity settings label the requested API input; measured speeds are retained separately in points/s. A quiet-stop control adds a one-physical-pixel slow finish before release.

Campaign	Gesture and sampling qualified	Total
Physical phone, public drag API	16	16
Physical phone, quiet stop	4	4
Simulator, public drag API	14	16
Simulator, quiet stop	3	4

Observation: the fast 100-point phone pulls continue outward after release by 28 points without a hold, and 33.33-39.67 points after an exact 400 ms hold. Slint adds zero outward travel. During the exact hold, UIKit's pan recognizer still reports a cached nonzero velocity. The quiet-stop control refreshes that estimate to about -1.336 points/s and adds no outward travel.

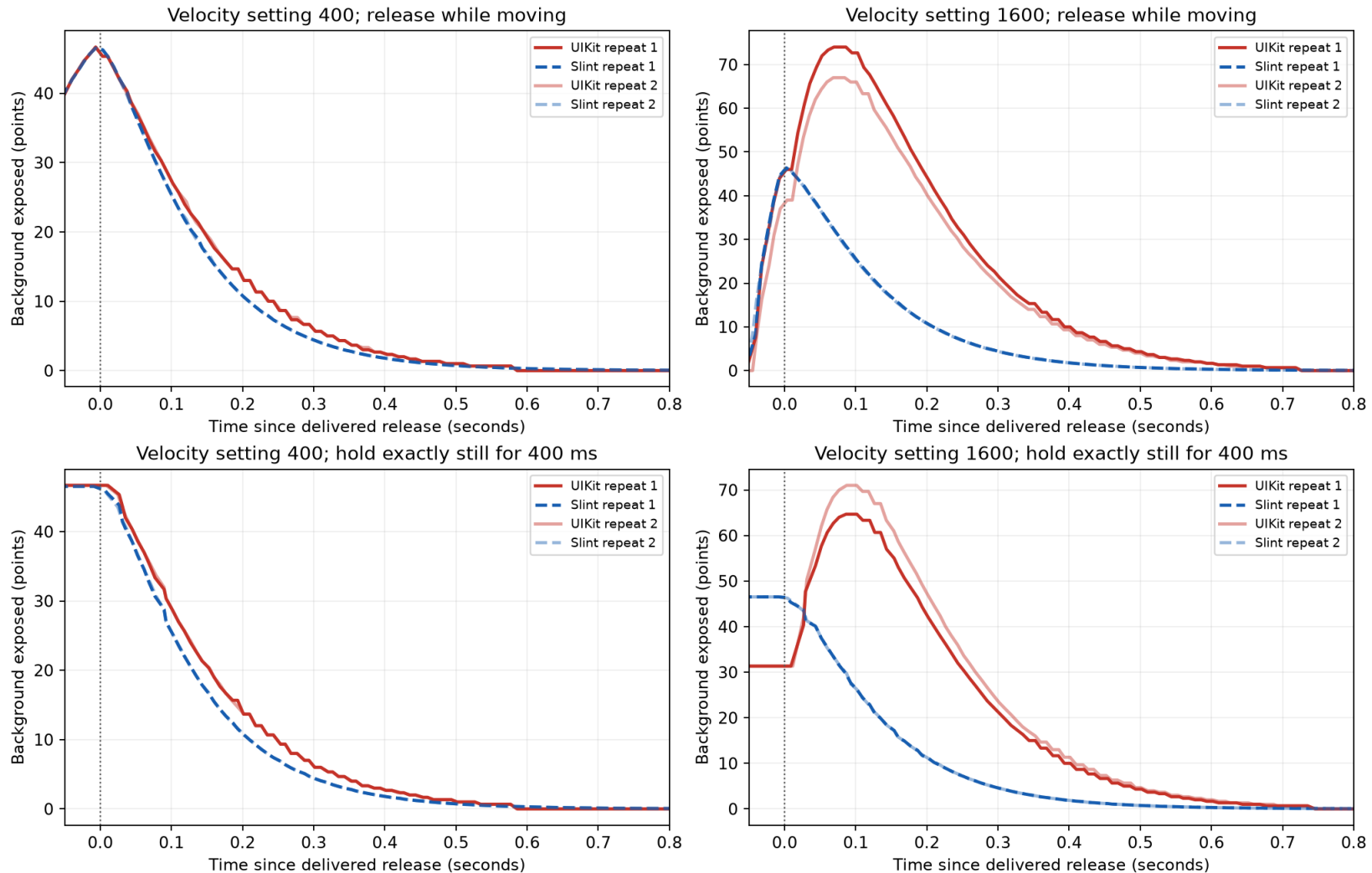
Slint's outside-bounds release calls `spring_back` without passing estimated release velocity. Its estimator also treats motion older than 40 ms as stopped. These source facts explain why one fixed return curve cannot cover the observed release states. They do not prove that UIKit changes spring coefficients: different initial velocities and nonlinear resistance can change the curve.

37/40 traces pass the predeclared 30 ms maximum sample-gap rule. Failed traces remain visible. Starting exposure sometimes differs because pan entry is unresolved, so return error includes that gap. A still hold is not a verified zero-velocity native spring. The earlier dense injector compressed requested timing and is excluded from this matrix.

Physical iPhone: Immediate release / exact hold, 100 points

UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

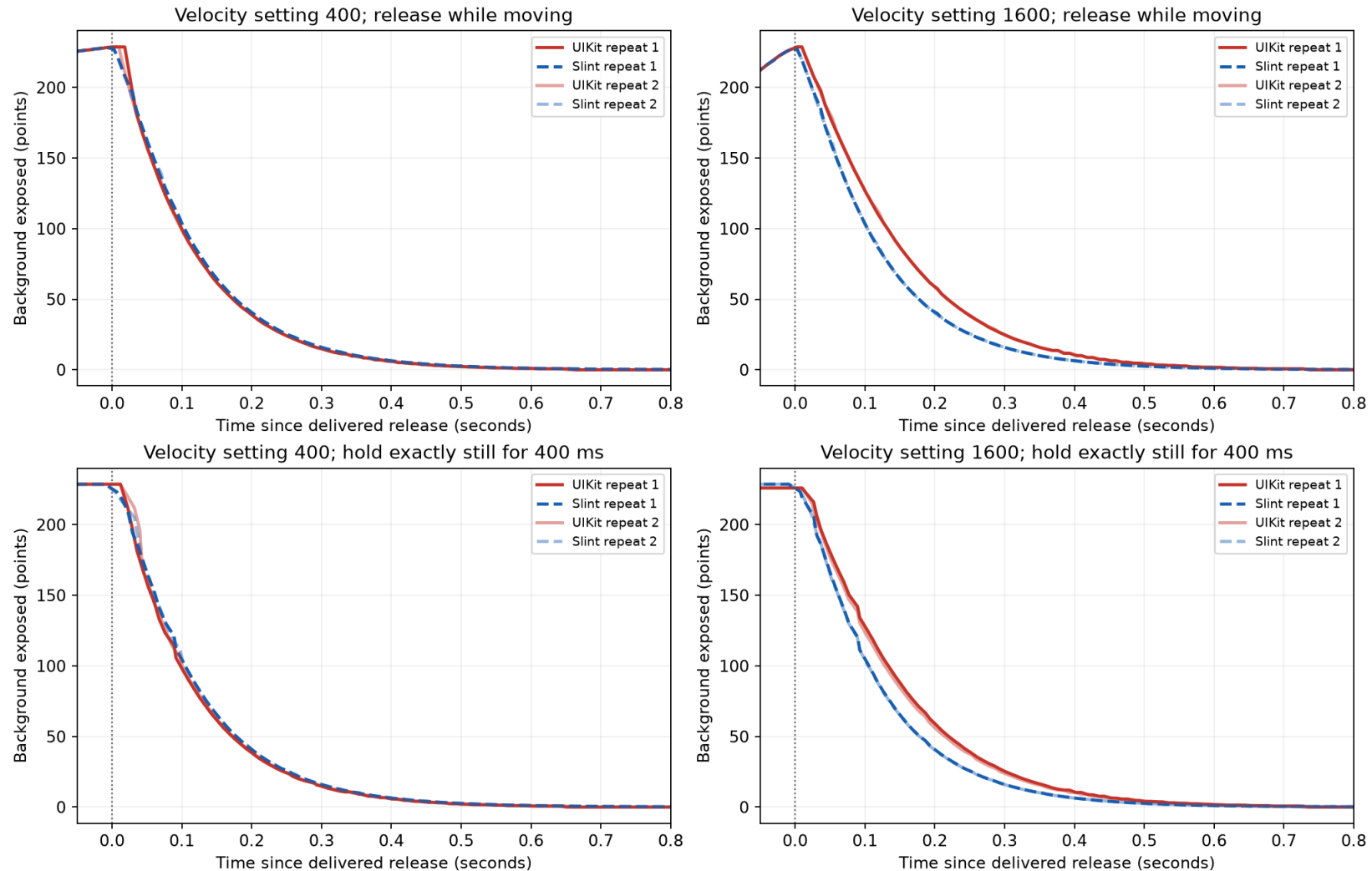
Phone: 100-point pull, moving versus exact hold



Physical iPhone: Immediate release / exact hold, 600 points

UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

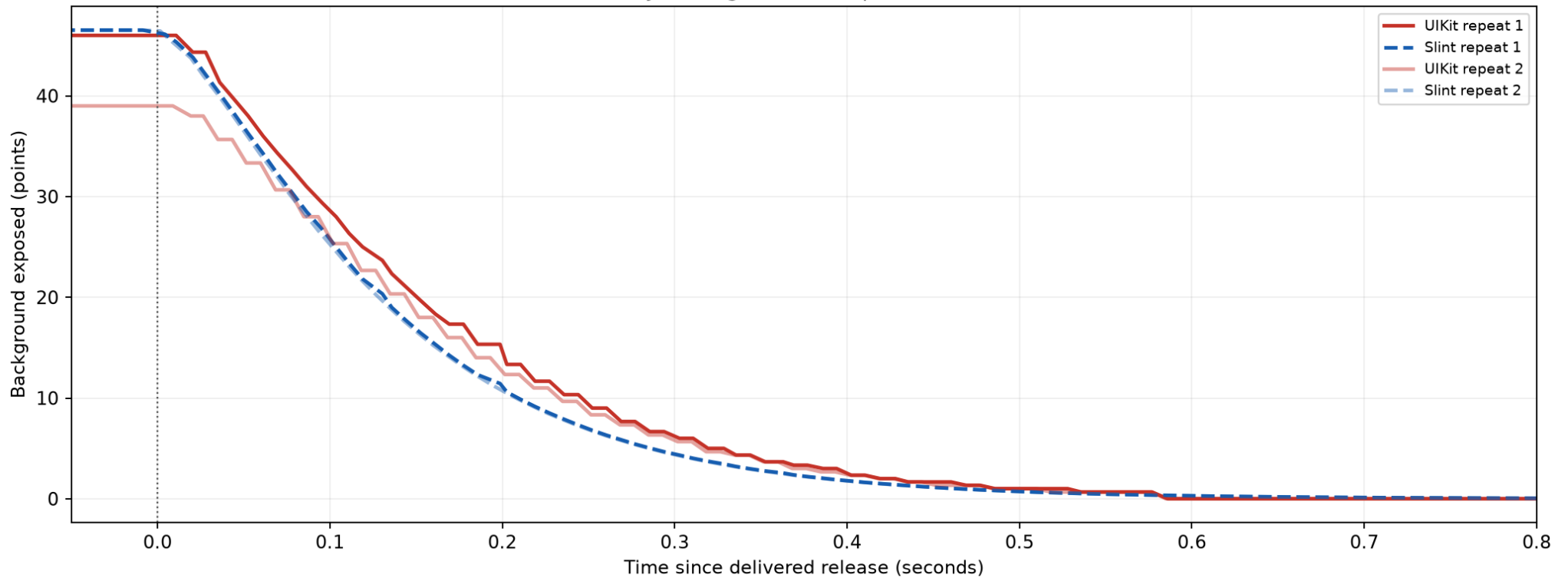
Phone: 600-point pull, moving versus exact hold



Physical iPhone: Quiet-stop control, 100 points

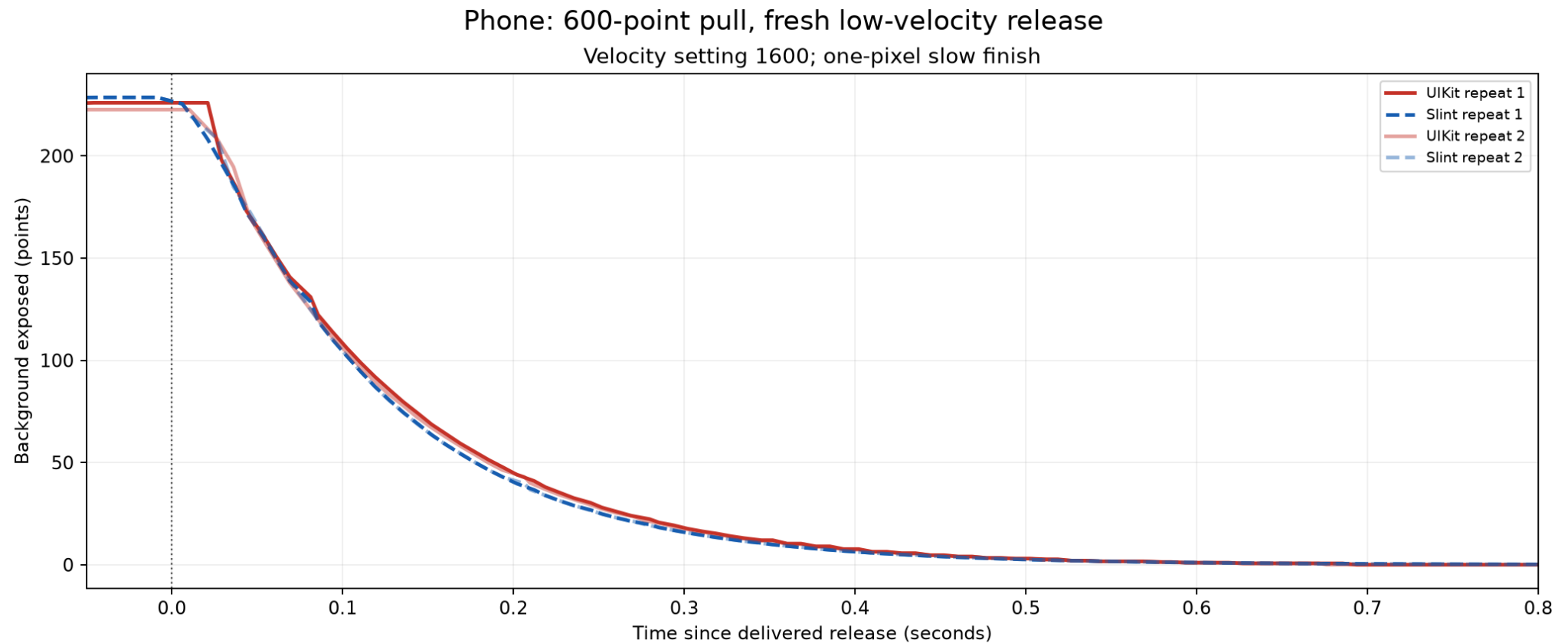
UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

Phone: 100-point pull, fresh low-velocity release
Velocity setting 1600; one-pixel slow finish



Physical iPhone: Quiet-stop control, 600 points

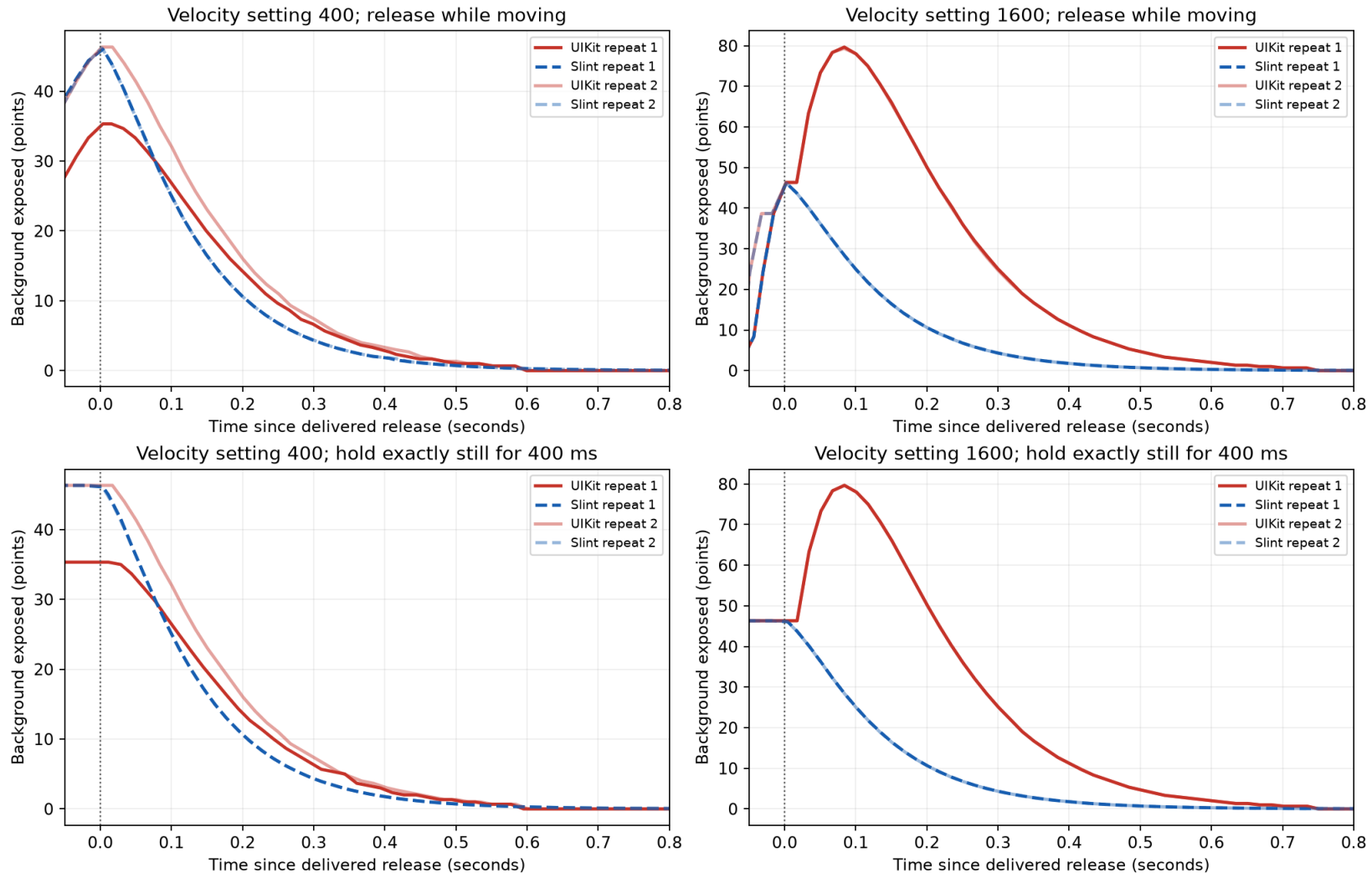
UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.



Simulator: Immediate release / exact hold, 100 points

UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

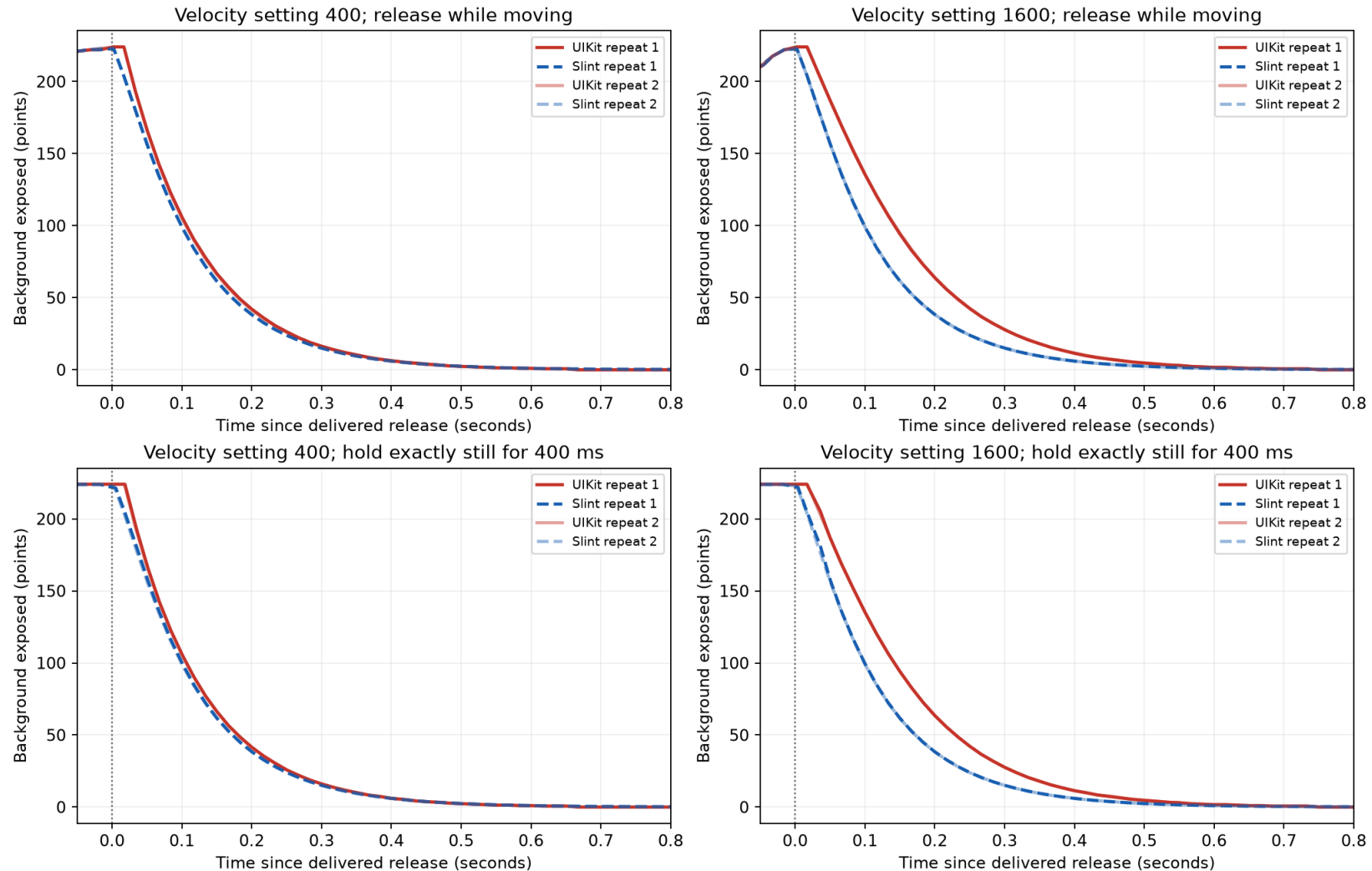
Simulator: 100-point pull, moving versus exact hold



Simulator: Immediate release / exact hold, 600 points

UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

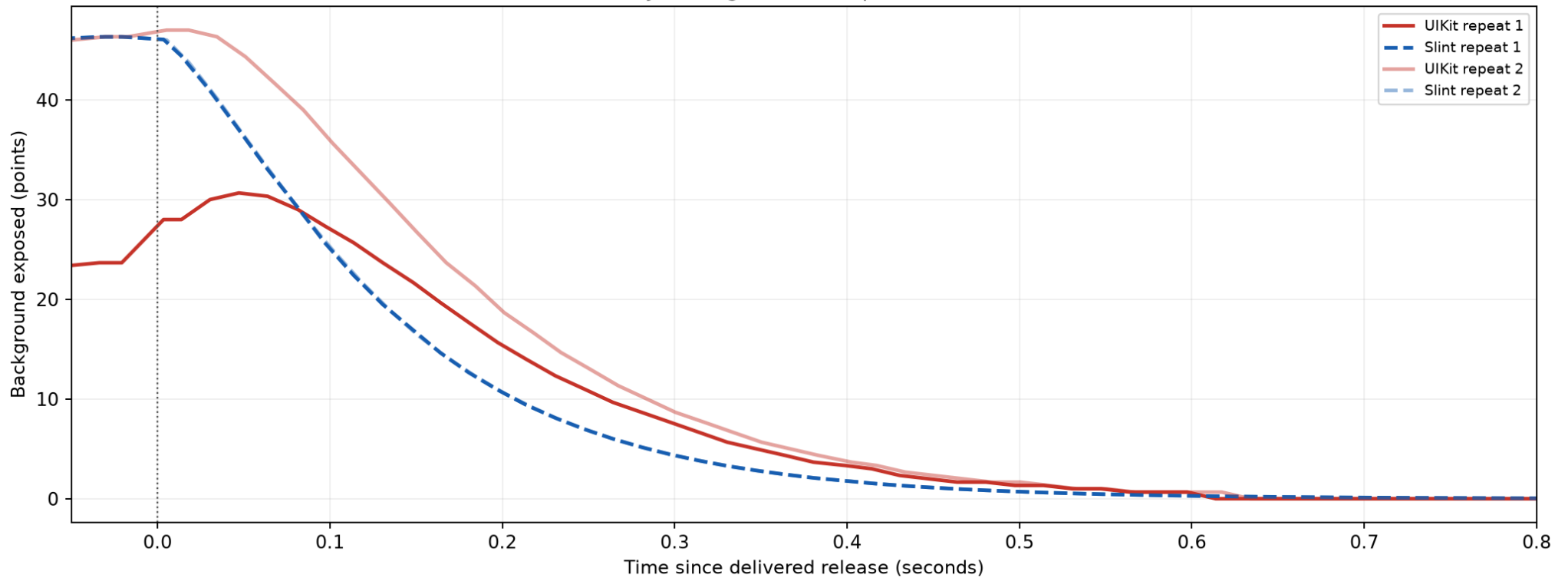
Simulator: 600-point pull, moving versus exact hold



Simulator: Quiet-stop control, 100 points

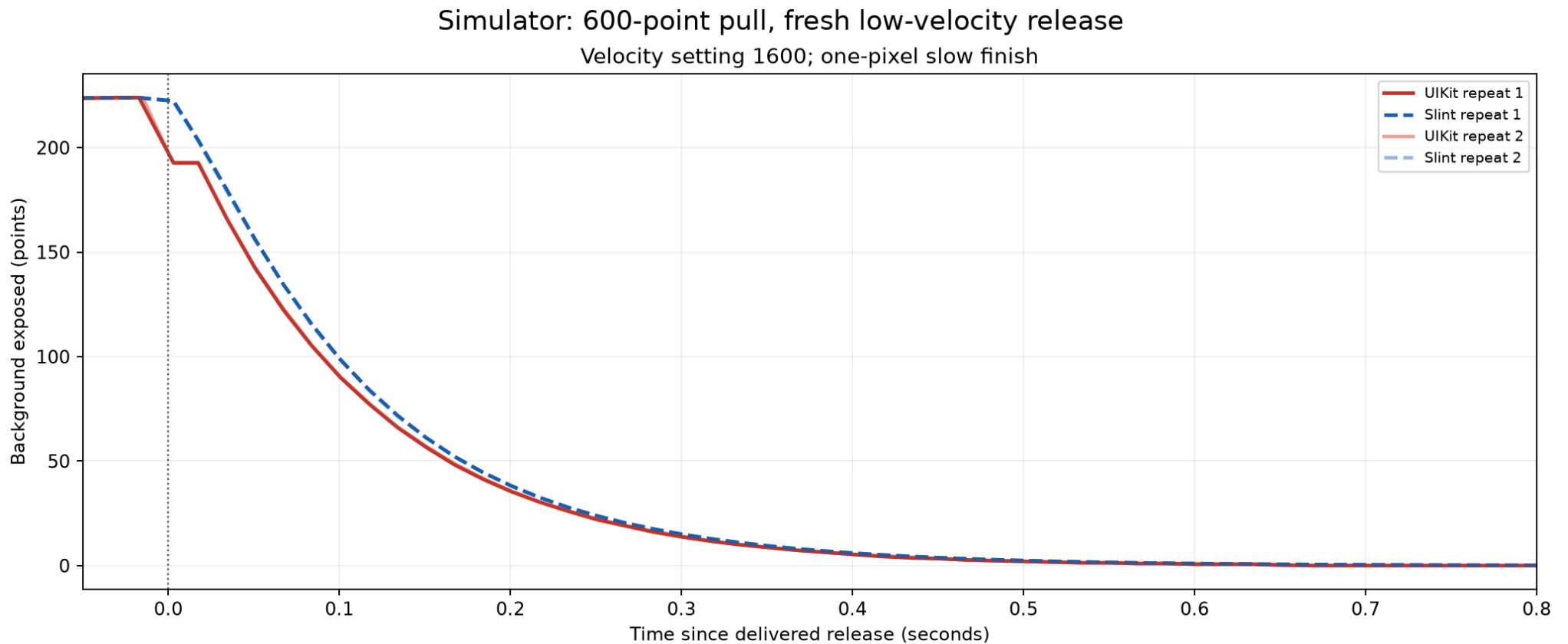
UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.

Simulator: 100-point pull, fresh low-velocity release
Velocity setting 1600; one-pixel slow finish



Simulator: Quiet-stop control, 600 points

UIKit red, Slint blue. Actual exposure and common release time; both repeats retained.



Hit the top after the finger lifts

Both lists start at item 10 (offset 648 points). A downward 200-point finger path moves toward the start of the full 1000-row list. Each of six velocity settings runs twice in a fresh launch. The finger must lift with both offsets still positive. The slow setting is a no-hit control.

Slint's in-bounds fling transfers measured model velocity into the existing collision spring. This is a separate path from releasing a finger already outside the bounds. The new test records natural free deceleration, the first edge crossing, peak background exposure, return, and settling. It does not copy native positions or velocities into Slint.

Result: all 12 phone traces and 11 of 12 simulator traces qualify. The simulator's first slow control has a 33.11 ms sampling gap and is excluded. Settings 1500 and above produce clear top collisions after release on both platforms.

In the eight phone collisions, Slint peaks earlier and settles 74-112 ms earlier than UIKit. Its fitted approach speed is higher, yet seven of eight peaks expose less background. The second 4500-setting repeat has nearly equal peaks but still different return timing. Both approach and collision response therefore need attention. Native phone release offsets range from 458 to 523 points; Slint releases at 458. The shared gesture does not force equal release poses or incoming velocities. An equal-state control is needed to isolate spring coefficients.

The simulator's repeated 3000 settings delivered substantially different behavior. Requested settings are not identical measured momentum. A 0.093-point Slint crossing at setting 1000 is retained as unresolved below the 0.5-point amplitude criterion. These observations do not identify UIKit's private spring constants or prove that they change.

Top-boundary collision measurements

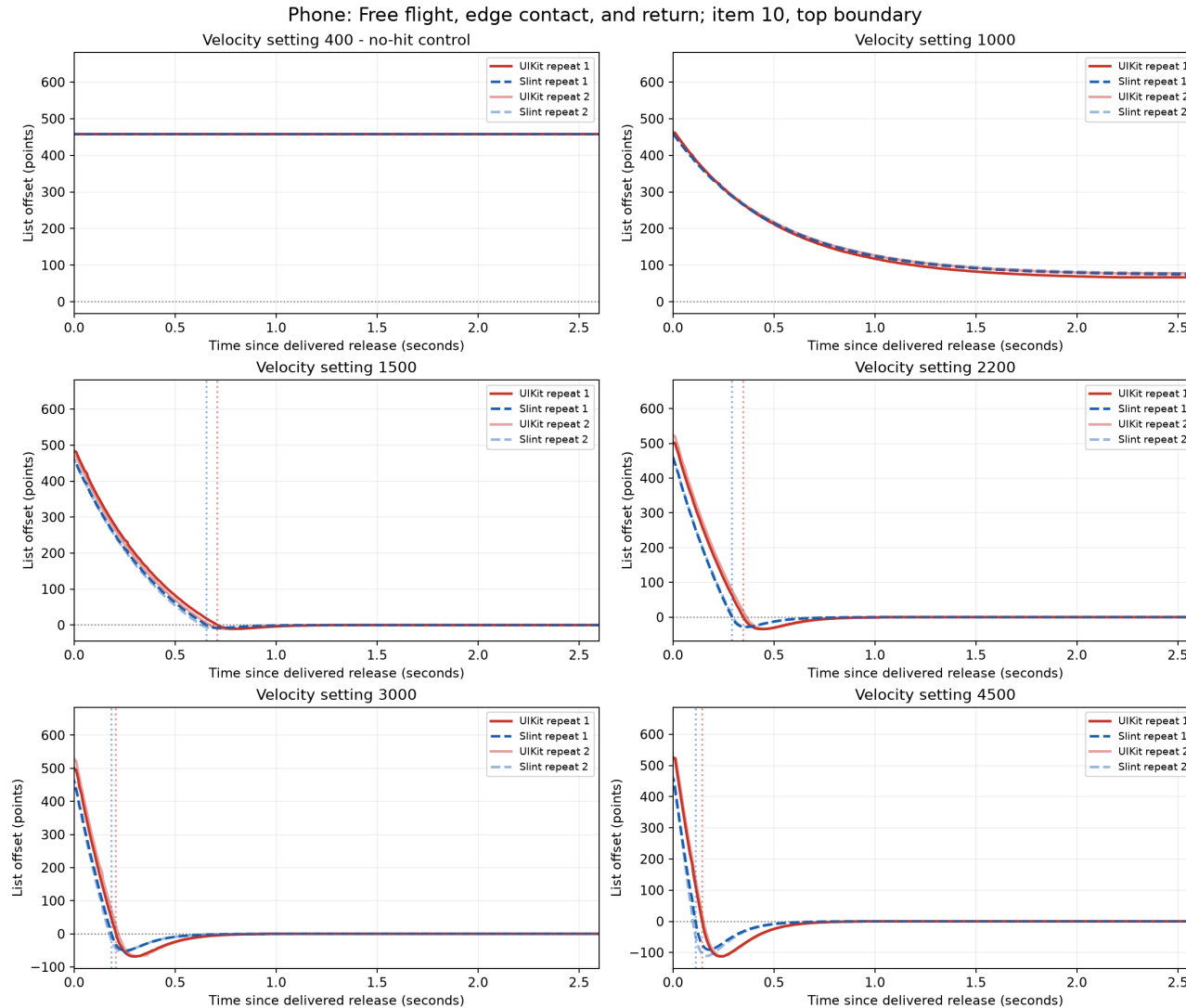
Ranges cover both repeats passing gesture and sampling guards; requested settings are not measured speeds.

Platform / setting	UIKit / Slint peak (pt)	UIKit / Slint settle (s)
phone / 400	no hit / no hit	no hit / no hit
phone / 1000	no hit / no hit	no hit / no hit
phone / 1500	10.33-11.33 / 8.28-8.88	0.55-0.55 / 0.44-0.45
phone / 2200	33.67-34.33 / 27.71-28.45	0.66-0.68 / 0.57-0.57
phone / 3000	67.00-68.33 / 50.64-54.63	0.74-0.75 / 0.64-0.65
phone / 4500	111.00-112.67 / 90.91-111.06	0.80-0.80 / 0.71-0.73
simulator / 400	no hit / no hit	no hit / no hit
simulator / 1000	no hit / 0.09-0.09	no hit / unresolved
simulator / 1500	20.33-20.67 / 14.81-14.87	0.61-0.61 / 0.51-0.51
simulator / 2200	43.00-43.33 / 31.39-31.79	0.70-0.70 / 0.60-0.60
simulator / 3000	23.33-80.00 / 23.74-121.81	0.63-0.77 / 0.57-0.75
simulator / 4500	47.33-47.33 / 34.09-34.88	0.70-0.70 / 0.60-0.60

Settling is measured from each observed impact to staying within 0.5 points. The charts retain one common release clock and do not align impacts. Sub-0.5-point edge excursions are marked unresolved in the evidence. Measured impact speeds and sample counts are in the aggregate JSON; missing speed estimates are explicit. In-bounds flings use the existing IOsFlick collision spring, not the outside-release spring candidate.

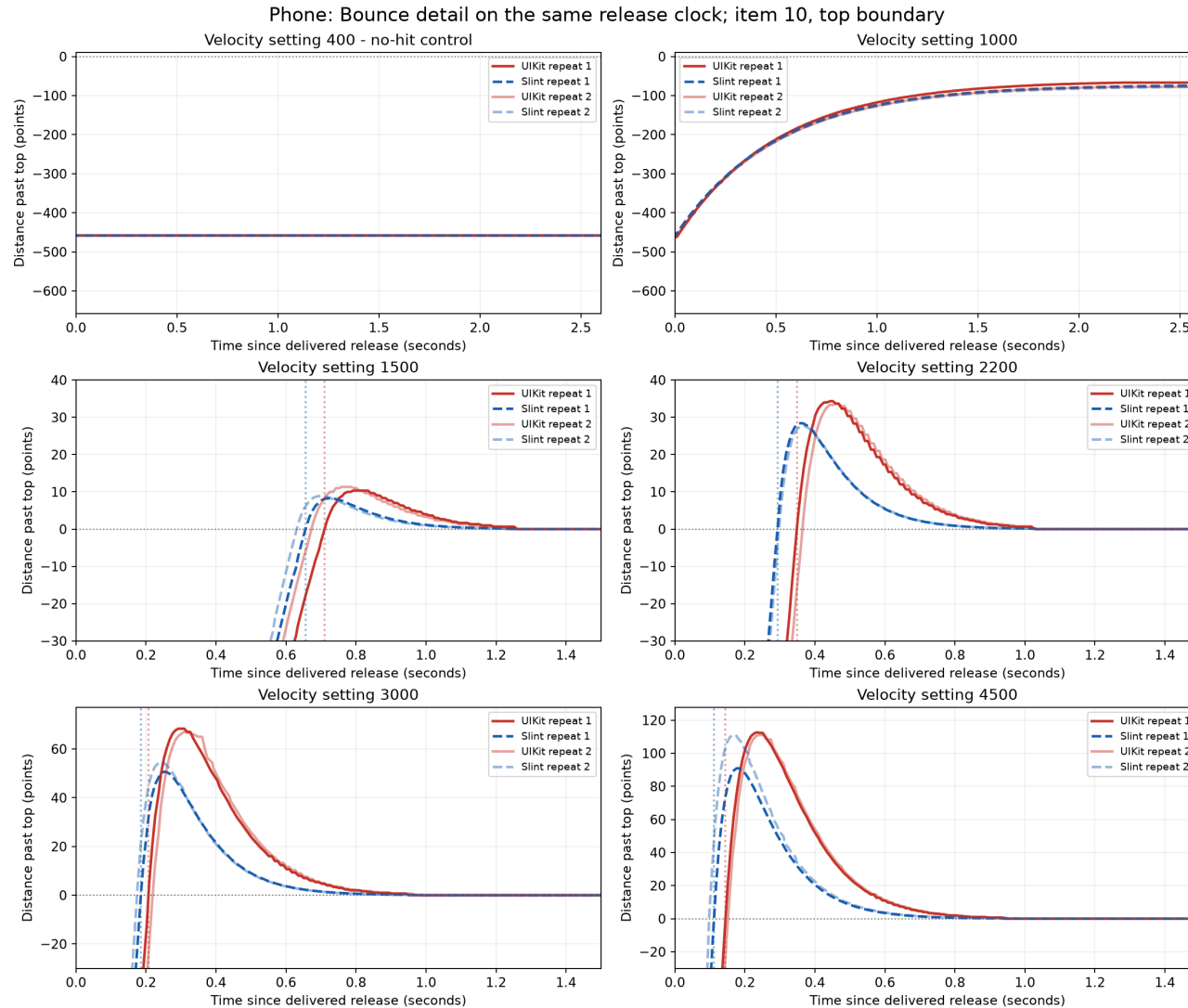
Physical iPhone: Free flight and top collision

Item 10, downward flick, finger up before impact. UIKit red, Slint blue; actual points and seconds.



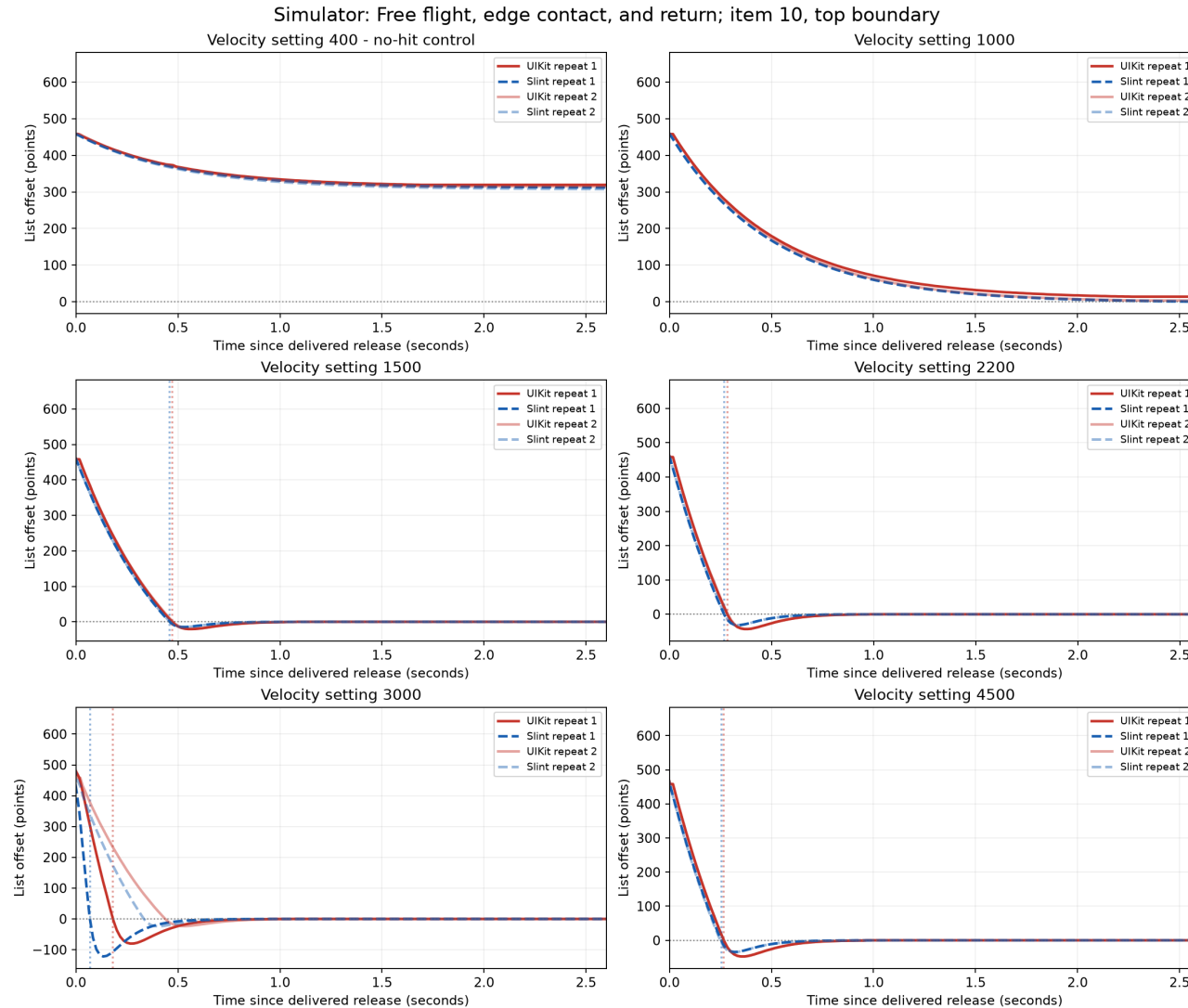
Physical iPhone: Top bounce detail

Item 10, downward flick, finger up before impact. UIKit red, Slint blue; actual points and seconds.



Simulator: Free flight and top collision

Item 10, downward flick, finger up before impact. UIKit red, Slint blue; actual points and seconds.



Simulator: Top bounce detail

Item 10, downward flick, finger up before impact. Uikit red, Slint blue; actual points and seconds.

